Introduction to Computer Programming

Fundamentals of C for Complete Beginners

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Outline

What is a Computer?

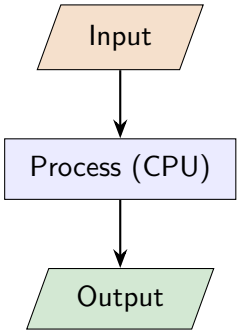
Definition

A **computer** is an electronic machine that accepts **data** (input), processes it according to a set of **instructions** (a program), and produces useful **information** (output).

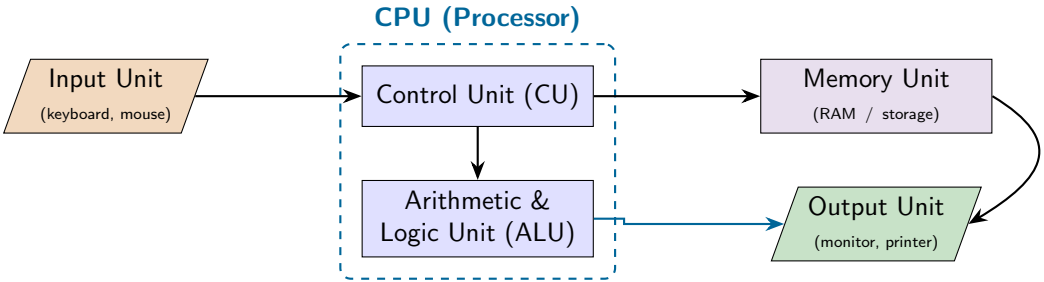
- It follows the simple cycle:

$$\textbf{INPUT} \rightarrow \textbf{PROCESS} \rightarrow \textbf{OUTPUT}$$

- It cannot "think" — it only does exactly what the program tells it, very fast and without mistakes.
- *Programming* is the art of writing those instructions.



Computer Organization: The Building Blocks



- **CPU** = "brain". **CU** directs traffic; **ALU** does the maths & comparisons.
- **Memory** temporarily holds data & instructions the CPU is working on.

Hardware vs. Software & Types of Memory

Hardware

The physical parts you can *touch* — CPU, RAM, hard disk, keyboard, monitor.

Software

The *programs* (instructions) that tell the hardware what to do — e.g. Windows, a game, a C program.

Primary Memory (fast, temporary)

- **RAM** — working memory, lost when power is off (volatile).
- **ROM** — read-only, permanent start-up instructions.

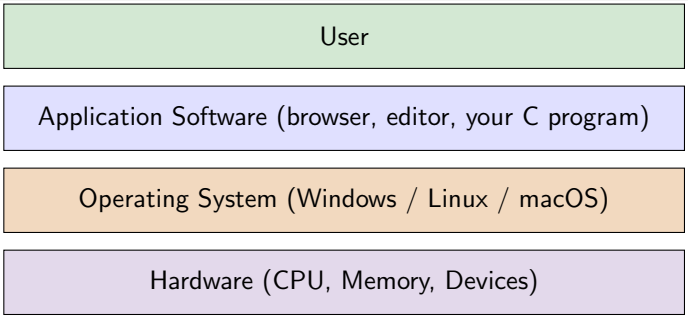
Secondary Memory (slow, permanent)

- Hard disk, SSD, USB drive — stores files even without power.

What is an Operating System (OS)?

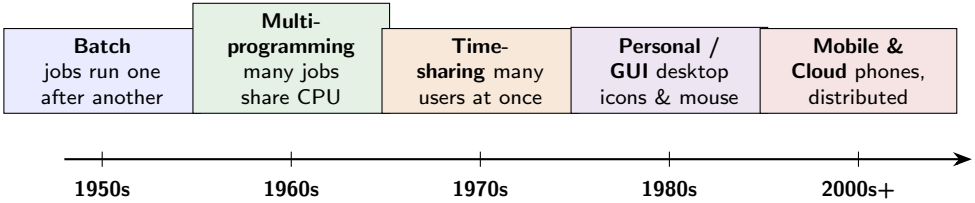
Operating System

An OS is the **master software** that manages the hardware and gives other programs a place to run. It is the bridge between **you**, the **programs**, and the **hardware**.



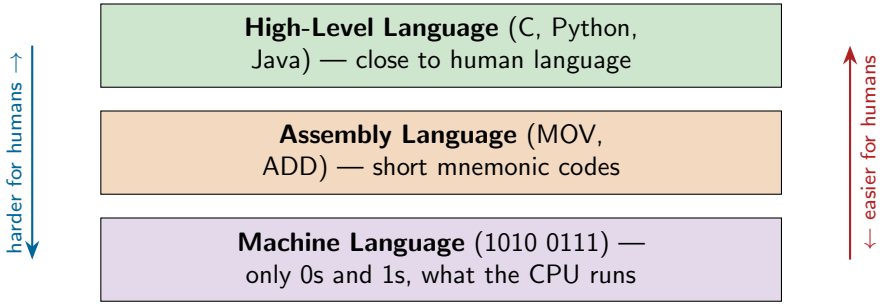
Examples of tasks: managing memory, running programs, handling files, controlling devices.

Evolution of Operating Systems: A Timeline



- Each stage made the computer **easier to use** and able to do **more at once**.
- Modern OSes let many programs and users run *simultaneously* on the same machine.

Levels of Programming Languages



- The CPU understands **only** machine language (binary).
- We write in a high-level language; a **translator** converts it to machine code.

Comparing the Three Levels

Feature	Human friendliness	Example (add 2 numbers)
Machine	Very hard (all 0/1)	`10110000 01100001`
Assembly	Hard (mnemonics)	`MOV AL, 61h ADD AL, BL`
High-level	Easy (English-like)	`c = a + b;`

Key idea

High-level languages like **C** let us write *one* readable line instead of many cryptic machine instructions — and the same C program can run on many different machines.

Key Software & Hardware Trends

Hardware trends

- **Moore's Law**: transistor count roughly doubles ∼every 2 years → faster, cheaper chips.
- Multi-core CPUs, GPUs.
- Smaller & mobile: phones, embedded, IoT.
- Cloud data-centres provide computing "on tap".

Software trends

- Open-source software & reusable libraries.
- Mobile apps and web applications.
- Artificial Intelligence & data science.
- Higher-level, safer languages built *on top of* C.

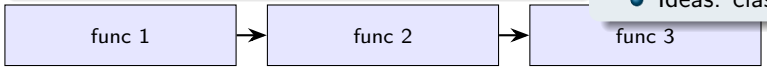
Hardware gets faster and cheaper; software gets smarter and more connected — and **C** sits underneath much of it.

Procedural vs. Object-Oriented Programming

Procedural (e.g. C)

Program = a sequence of **procedures / functions** that operate on data.

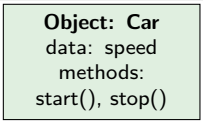
- Think: a *recipe* — do step 1, then 2, then 3.
- Data and functions are separate.



Object-Oriented (e.g. C++, Java)

Program = a set of **objects** that bundle data *and* the functions acting on it.

- Think: real-world *objects* (a Car that can `start()`, `stop()`).
- Ideas: class, object, inheritance.



In this course we focus on the **procedural** style using **C**.

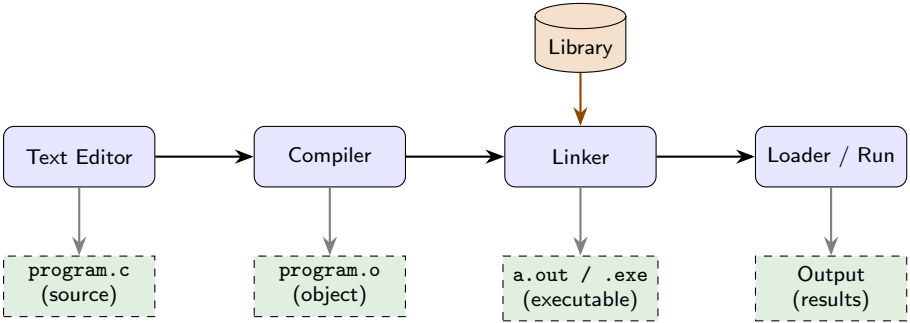
Why C?

- Created by **Dennis Ritchie** at Bell Labs (early 1970s).
- **Powerful** yet **small** — close to the hardware but readable.
- The "mother language": Linux, databases, and parts of Windows are written in C.
- Learning C teaches you how a computer *really* works.

C is a great first language because

- It is **fast**.
- It is used **everywhere**.
- Its ideas carry over to C++, Java, Python, ...

From Source Code to Running Program



One command does it all (using GCC)

```
gcc program.c -o program
then   ./program
```

Each step's output feeds the next. If the compiler reports **errors**, fix them and compile again — this "edit → compile → run" loop is normal!

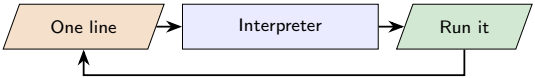
Translators: Compiler vs. Interpreter

Why we need a translator

A **translator** converts our high-level C into machine code. The two main kinds are the **compiler** and the **interpreter**.

Compiler (C uses this)

Interpreter (e.g. Python)

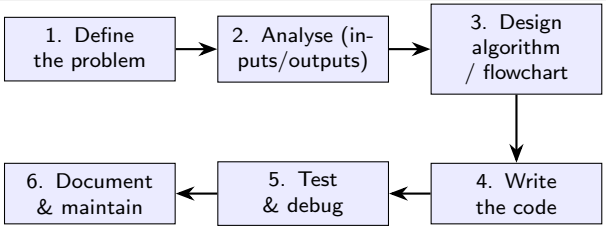


Compiler	**Interpreter**
Translates the *whole* program at once	Translates *line by line*
Produces a separate executable file	No separate executable file
Faster; all errors reported together	Slower; errors reported line-wise

Steps in Problem Solving

Before writing code, plan the solution

Good programmers think first, then type. Solving a problem on a computer follows a clear set of steps.



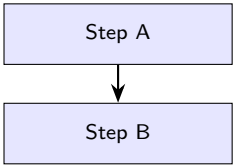
- **Analysis** asks: what are the *inputs*, the *outputs*, and the *constraints*?
- Only step 4 is actual coding — most of the thinking happens *before* it.

Structured Programming

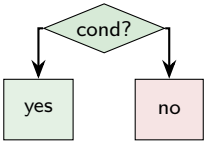
Idea

Build every program from just **three** basic control structures. This keeps code clear, correct, and easy to read (no "spaghetti" jumps).

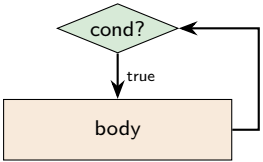
1. Sequence
(do in order)



2. Selection
(choose a path)



3. Repetition
(loop)



Algorithm and Pseudo-code

Algorithm

A finite, step-by-step procedure to solve a problem — *before* writing any code.

Pseudo-code

An informal, English-like sketch of the code — language independent.

Algorithm: add two numbers

1. Start
2. Read two numbers `a` and `b`
3. Compute `sum = a + b`
4. Display `sum`
5. Stop

```
BEGIN
   INPUT a, b
   sum ← a + b
   OUTPUT sum
END
```

Good pseudo-code translates almost line-by-line into C.

Characteristics of a Good Algorithm

- **Input** — zero or more well-defined inputs.
- **Output** — at least one useful output.
- **Definiteness** — every step is precise and unambiguous.
- **Finiteness** — it must end after a finite number of steps.
- **Effectiveness** — each step is basic enough to be carried out.

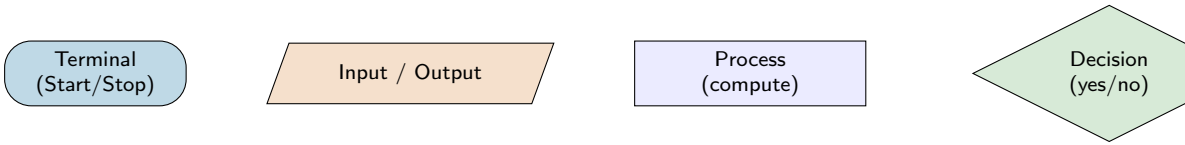
Rule of thumb

If a stranger can follow your steps and get the *same* correct answer every time, it is a good algorithm.

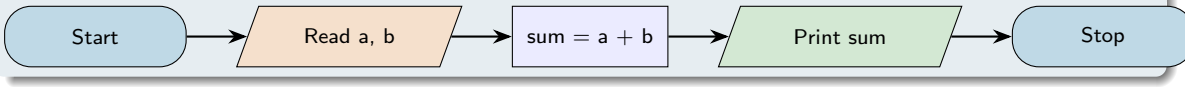
Why it matters

A clear algorithm makes the C code almost write itself — and makes bugs easy to spot.

Flowchart Symbols You Should Know



Flowchart: add two numbers



The Skeleton of Every C Program

```c
#include <stdio.h>    /* 1: header */

int main(void)        /* 2: start  */
{                     /* 3: body   */
    /* your statements go here */

    return 0;         /* 4: exit   */
}
```

1. **Preprocessor directive** — `#include` pulls in a library (here `stdio.h` for input/output).
2. `main()` — execution always begins here; every program has exactly one.
3. **Body** — the statements between { }; each ends with a semicolon ;
4. `return 0;` — tells the OS the program finished successfully.

Learn this shape first — every program we write just fills in the body.

The C Character Set and Tokens

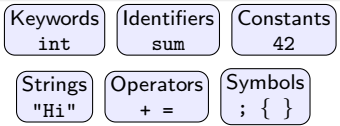
Character set

The symbols C understands:

- Letters `A--Z`, `a--z`
- Digits `0--9`
- Special symbols `+ - * / = % ; , ...`
- Whitespace (space, tab, newline)

Tokens — the smallest units

Characters group into **tokens**:



A compiler first breaks your code into these tokens.

Keywords: C's Reserved Words

What is a keyword?

A **keyword** has a fixed, special meaning in C. There are **32** of them — you cannot use them as your own names. C is **case-sensitive**: `int` is a keyword, `INT` is not.

`auto`	`break`	`case`	`char`	`const`	`continue`	`default`	`do`
`double`	`else`	`enum`	`extern`	`float`	`for`	`goto`	`if`
`int`	`long`	`register`	`return`	`short`	`signed`	`sizeof`	`static`
`struct`	`switch`	`typedef`	`union`	`unsigned`	`void`	`volatile`	`while`

The ones we use most as beginners: `int`, `float`, `char`, `if`, `else`, `for`, `while`, `return`.

Identifiers: Naming Your Variables

Identifier

The **name** you give to a variable, function, or constant.

Rules

- Use letters, digits, and underscore `_`.
- Must *start* with a letter or `_` (not a digit).
- No spaces and no special symbols.
- Cannot be a keyword.
- Case-sensitive: age $\neq$ `Age`.

Valid	Invalid
`total`	`2sum` (starts with digit)
`marks1`	`my age` (has a space)
`_count`	`float` (keyword)
`avgScore`	`rate%` (bad symbol)

Tip: choose *meaningful* names like `total`, not `t`.

The C Standard Library

What is it?

A ready-made collection of **functions** that come with C, so you don't reinvent the wheel. You get access to them with `#include <header.h>`.

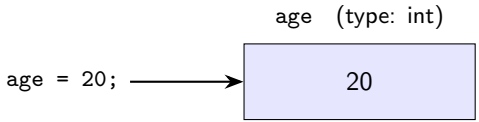
Header	Purpose	Example functions
`stdio.h`	input / output	`printf`, `scanf`, `getchar`
`stdlib.h`	general utilities	`malloc`, `rand`, `abs`, `atoi`
`math.h`	mathematics	`sqrt`, `pow`, `sin`, `ceil`
`string.h`	text handling	`strlen`, `strcpy`, `strcmp`
`ctype.h`	character tests	`isdigit`, `toupper`

e.g. `#include <math.h>` then `y = sqrt(25.0);` gives `y = 5.0`.

Variables and Data Types

Variable

A **named box** in memory that stores a value. You must declare its *type* first: `int age = 20;`



- The **type** decides what values fit and how much memory is used.
- Choose the smallest type that comfortably holds your data.

The Basic Data Types

Type	Stores	Typical size	Example
`int`	whole numbers	4 bytes	`int n = 42;`
`float`	real numbers (decimals)	4 bytes	`float g = 9.8f;`
`double`	real numbers (more precise)	8 bytes	`double pi = 3.14159;`
`char`	a single character	1 byte	`char grade = 'A';`

Modifiers

`short`, `long`, `unsigned` adjust range/size, e.g. `unsigned int` (no negatives), `long int` (bigger).

A character is a small number

`'A'` is stored as the integer 65 (its ASCII code).

Size and Range of Each Variable

Every type occupies a fixed amount of memory

The **size** (in bytes) decides the **range** of values a variable can hold. Sizes are typical for a modern computer.

Type	Size	Bits	Approx. range	Specifier
`char`	1 byte	8	-128 to 127	`%c`
`int`	4 bytes	32	-2.1×10^9 to 2.1×10^9	`%d`
`float`	4 bytes	32	$\sim$6–7 significant digits	`%f`
`double`	8 bytes	64	$\sim$15 significant digits	`%lf`

Find the size yourself

The `sizeof` operator returns a type's size in bytes:

```c
printf("%d", sizeof(int));
// prints 4
```

Constants: Values That Never Change

Constant

A value that is fixed and **cannot be changed** while the program runs (e.g. π, the number of days in a week). Using constants makes code safer and easier to read.

Using the `const` keyword:

```
const float PI = 3.14159;
const int DAYS = 7;
```

Using `#define` (a macro):

```
#define MAX 100
#define PI 3.14159
```

- `const` makes a *typed* read-only variable; `#define` does a plain text substitution before compiling.
- By convention, constants are written in `UPPERCASE`. Trying to change one is an error.

Variable Scope: Local vs. Global

```c
int x = 5;          /* global */

int main(void) {
    int y = 10;    /* local  */
    printf("%d %d", x, y);
    return 0;
}
```

Scope = where a variable is visible

- **Local** — declared *inside* a function; usable only there.
- **Global** — declared *outside* all functions; usable everywhere.

Prefer local variables — they keep parts of the program independent and reduce accidental bugs.

Arithmetic Operators

Operator	Meaning	Example	Result
+	addition	`7 + 3`	10
-	subtraction	`7 - 3`	4
*	multiplication	`7 * 3`	21
/	division	`7 / 3`	2 (integer!)
%	modulus (remainder)	`7 % 3`	1

Beginner trap: integer division

`7 / 3` gives 2, *not* 2.33, because both are integers. Use `7.0 / 3` to get `2.333...` The `%` operator works only on integers.

Operator Precedence (Order of Operations)

- C follows maths-like precedence:

  `*` `/` `%` **before** `+` `-`

  - Same level → left to right.
- Use **parentheses** to be explicit and clear.

Example

```
2 + 3 * 4   →   2 + 12   →   14
(2 + 3) * 4   →   20
```

Assignment & shortcuts

- `x = 5;` stores 5 in `x`.
- `x += 2;` means `x = x + 2;`
- `x++;` increases `x` by 1.
- `x--;` decreases `x` by 1.

Relational and Logical Operators

Relational — compare two values

Result is **true (1)** or **false (0)**.

Op	Meaning	Example
==	equal to	5 == 5 → 1
!=	not equal	5 != 3 → 1
>	greater than	5 > 3 → 1
<	less than	5 < 3 → 0
>=	greater/equal	5 >= 5 → 1
<=	less/equal	4 <= 3 → 0

Logical — combine conditions

Used to join or invert true/false tests.

Op	Meaning	True when
&&	AND	both are true
||	OR	at least one is true
!	NOT	flips true/false

e.g. (age >= 18 && age < 60) is true only for ages 18 to 59.

Assignment and Increment Operators

Operator	Meaning
`x = 5`	assign 5 to x
`x += 2`	`x = x + 2`
`x -= 3`	`x = x - 3`
`x *= 4`	`x = x * 4`
`x /= 5`	`x = x / 5`

```
1 int x = 10;
2 x += 5;   // 15
3 x -= 3;   // 12
4 x *= 2;   // 24
5 x /= 4;   // 6
```

Final value of `x` is $\boxed{6}$.

A Peek at Bitwise Operators

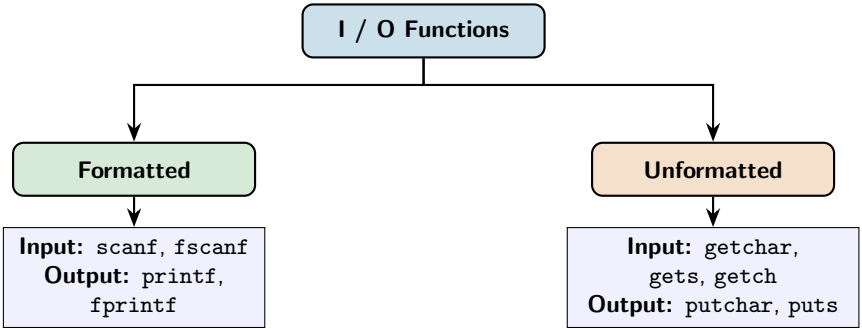
Working at the level of bits

Bitwise operators act on the individual **0/1 bits** of an integer. They are common in *embedded systems* and *hardware control* — handy to recognise even as a beginner.

```c
int a = 5;   // 0101
int b = 3;   // 0011

a & b;   // 0001 = 1  (AND)
a | b;   // 0111 = 7  (OR)
a ^ b;   // 0110 = 6  (XOR)
~a;      // flips all bits
a << 1;  // 1010 = 10 (x2)
a >> 1;  // 0010 = 2  (/2)
```

- & AND — 1 only if *both* bits are 1.
- | OR — 1 if *any* bit is 1.
- ^ XOR — 1 if the bits *differ*.
- ~ NOT — flips every bit.
- << left shift — multiplies by 2.
- >> right shift — divides by 2.

Classification of I/O Functions



- **Formatted** functions use *format specifiers* (%d, %f, ...) to control layout — our focus.
- **Unformatted** functions read/write raw characters or strings without formatting.

Output with `printf` — Format Specifiers

Specifier	Used for	Example
`%d`	integer	`printf("%d", 42);` → `42`
`%f`	float / double	`printf("%f", 3.14);` → `3.140000`
`%c`	single character	`printf("%c", 'A');` → `A`
`%s`	string (text)	`printf("%s", "Hi");` → `Hi`

Width & precision

`%5d` — at least 5 columns wide.
`%.2f` — 2 digits after the point.
`printf("%.2f", 3.14159);` → `3.14`

Escape sequences

`\n` new line `\t` tab
`\\` backslash `\"` quote

Input with scanf

```c
int age;
printf("Enter your age: ");
scanf("%d", &age);          /* note the & (address of age) */
printf("Next year you will be %d\n", age + 1);
```

- `scanf` reads typed input and stores it in a variable.
- **Crucial:** put an `&` ("address of") before the variable — `&age`.
- The specifier must match the type: `%d` int, `%f` float, `%c` char, `%s` string.

Reading several values

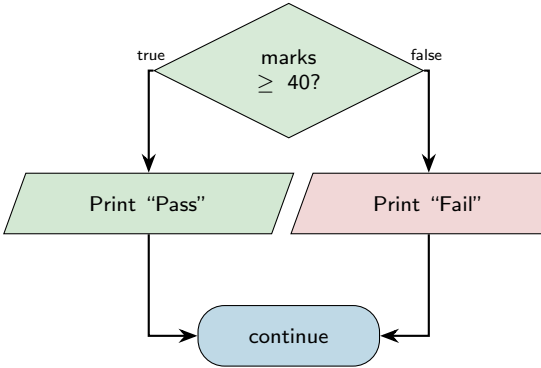
`scanf("%d %f", &n, &x);` reads an integer then a float.

Sample run: `Enter your age: 20` → `Next year you will be 21`

Making Decisions: `if` / `if-else`

```c
if (marks >= 40) {
    printf("Pass\n");
} else {
    printf("Fail\n");
}
```

- The condition in ( ) is either **true** or **false**.
- Comparisons: == != < > <= >=
- **Trap:** == tests equality; = *assigns*!



Choosing Among Many: `switch`

```c
switch (choice) {
    case 1:
        printf("Add\n");
        break;
    case 2:
        printf("Subtract\n");
        break;
    default:
        printf("Invalid\n");
}
```

- Compares one variable against several `case` values.
- `break;` stops "falling through" to the next case — **don't forget it**.
- `default:` runs if nothing else matches.
- Cleaner than a long `if--else if` chain.

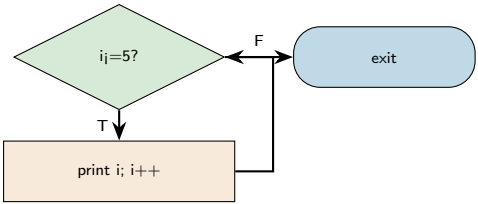
Note

`case` labels must be constant integers or characters (e.g. `1`, `'A'`).

Loops: `while` and `do--while`

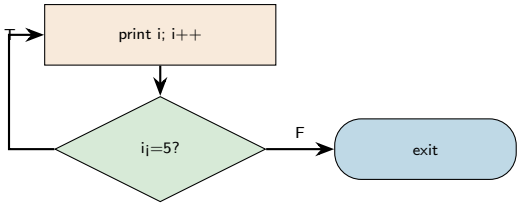
`while` — test *first*, may run 0 times:

```c
int i = 1;
while (i <= 5) {
    printf("%d ", i);
    i++;
}
```



`do--while` — test *after*, runs ≥ 1 time:

```c
int i = 1;
do {
    printf("%d ", i);
    i++;
} while (i <= 5);
```



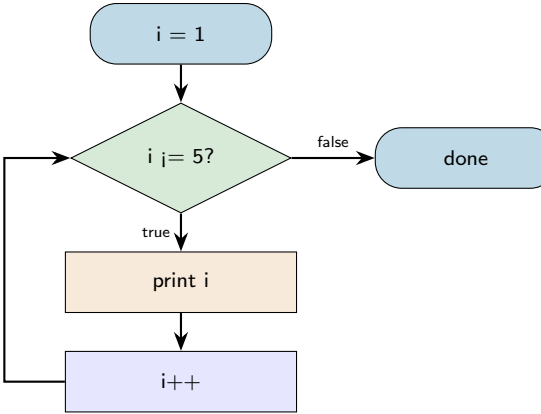
Both print `1 2 3 4 5`. Use `do--while` when the body must run at least once.

Loops: the `for` Loop

```c
for (i = 1; i <= 5; i++) {
    printf("%d ", i);
}
```

The three parts of a `for` loop:

1. **Init**: `i = 1` — runs once at the start.
2. **Condition**: `i <= 5` — checked before each pass.
3. **Update**: `i++` — runs after each pass.



A `for` loop is a compact `while` loop — best when you know the count in advance.

break and continue

`break` — exit the loop immediately.

```c
for (i = 1; i <= 10; i++) {
    if (i == 4) break;
    printf("%d ", i);
}
// prints: 1 2 3
```

`continue` — skip to the next iteration.

```c
for (i = 1; i <= 5; i++) {
    if (i == 3) continue;
    printf("%d ", i);
}
// prints: 1 2 4 5
```

Remember

`break` **leaves** the loop; `continue` **jumps to the next round**. Both also appear inside `while`/`do--while`; `break` additionally ends a `switch` case.

Your First C Program: "Hello, World!"

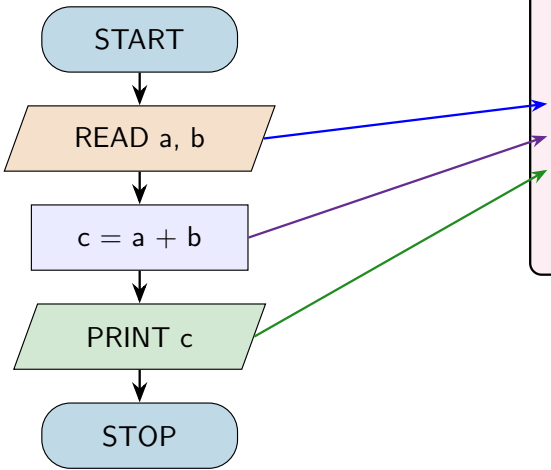
```c
#include <stdio.h>          /* bring in input/output library */

int main(void)              /* execution starts here */
{
    printf("Hello, World!\n");   /* print a message */
    return 0;                    /* 0 = success */
}
```

- We now know the skeleton, keywords, data types and I/O — so we can write real programs.
- `printf` prints text; `\n` starts a new line.

Output: `Hello, World!`

- Every statement ends with a semicolon `;`
- `return 0;` reports success to the OS.

Worked Example 1: Adding Two Numbers

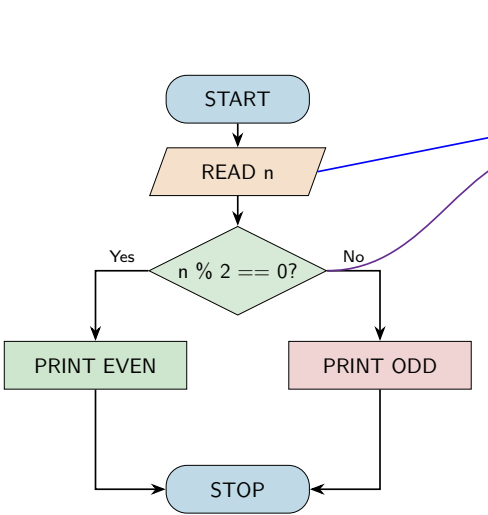



```c
#include <stdio.h>
int main(void)
{
    int a, b, c;
    scanf("%d %d", &a, &b);
    c = a + b;
    printf("%d", c);
    return 0;
}
```

Variable Declaration

Each flowchart step on the left maps to a coloured line of C code on the right.

Worked Example 2: Even or Odd (`if--else`)




```c
#include <stdio.h>
int main(void)
{
    int n;
    scanf("%d", &n);
    if (n % 2 == 0)
        printf("EVEN");
    else
        printf("ODD");
    return 0;
}
```

Variable Declaration

The green and red branches match the two `printf` lines. A number is even when `n % 2` is 0.

Worked Example 3 — Sum of 1 to N (a `for` loop)

```c
#include <stdio.h>

int main(void)
{
    int n, i, sum = 0;

    printf("Enter N: ");
    scanf("%d", &n);

    for (i = 1; i <= n; i++)
        sum += i;    /* sum = sum + i */

    printf("Sum 1..%d = %d\n", n, sum);
    return 0;
}
```

Trace for N = 5

i	sum
1	1
2	3
3	6
4	10
5	15

Output:

```
Sum 1..5 = 15
```

Worked Example 4 — Simple Calculator (`switch`)

```c
#include <stdio.h>
int main(void) {
    float a, b;  char op;
    printf("Enter: a op b  (e.g. 6 * 7): ");
    scanf("%f %c %f", &a, &op, &b);

    switch (op) {
        case '+': printf("= %.2f\n", a + b); break;
        case '-': printf("= %.2f\n", a - b); break;
        case '*': printf("= %.2f\n", a * b); break;
        case '/':
            if (b != 0) printf("= %.2f\n", a / b);
            else        printf("Cannot divide by zero!\n");
            break;
        default:  printf("Unknown operator\n");
    }
    return 0;
}
```

Summary & What Comes Next

You now know

- How a computer is organized and what an OS does.
- Machine / assembly / high-level languages.
- How a C program is written, compiled, and run.
- Data types, operators, and control structures.
- Formatted input/output with `printf`/`scanf`.

Practice makes perfect

- Type and run every example yourself.
- Deliberately make errors — read the messages.
- Modify examples: change values, add features.

Coming up

Functions, arrays, strings, and pointers.

Thank You

Questions?